µPolar : An Interactive 2D Visualization Tool for Microscopic Time-series Images

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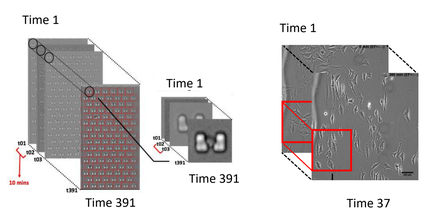
Summary

Time-lapse microscopy is an effective research tool to monitor cell behavior and cell divisions. Recent advances in microfluidics have accelerated the adoption of time-lapse microscopy in research. However, it is challenging to visualize and interpret the time-series data gathered through time-lapse microscopy. We have developed a circular plotting software tool, µPolar, to visualize the trends and patterns of the cell movements and cell division events in a time-series. µPolar is interactive and easy to use. We demonstrate the utility of µPolar by visualizing the events of dividing yeast cells where cell divisions lead to oscillating plotting patterns, and in migrating mouse fibroblasts where cell shapes change during the migration. µPolar potentially could be applied to other types of time-series of microscopic images.

Motivation

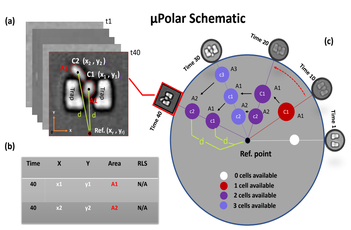
Time-lapse Images → **How to visualize :**

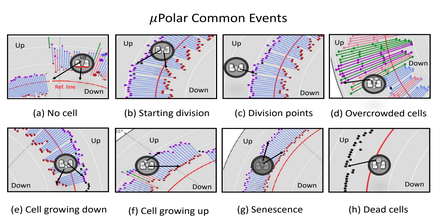


Yeast cells Microfluidic image | **Mouse fibroblast cells Microscopic images**

- **cell movements?**
- **cell size variation**
- **cell division?**
- **count cell replicative lifespan?**

Methods & Materials





(a) A sequence of microfluidic sub-images (60x60 pixels). The front image was labeled with cell numbers, coordinates, cell boundaries, and the reference point.

(b) A sample of µPolar dataset format with optional features of area and RLS, which correspond to the front image in a.

(c) µPolar plot with a representation of cell location on the plot and cell color tag. Black arrows track the same cell objects at subsequent time-points

(a) There is no cell at the presented time-point (white dot)

(b) The two cells represented by dark red dots and purple dots are close to each other, and division already happened

(c) The black dots represent a division after the cell completed its separation cycle

(d) Indication of overcrowded cell events

(e) Representation of steady cells in the up region (purple color) and developing cells in the down region (dark red color)

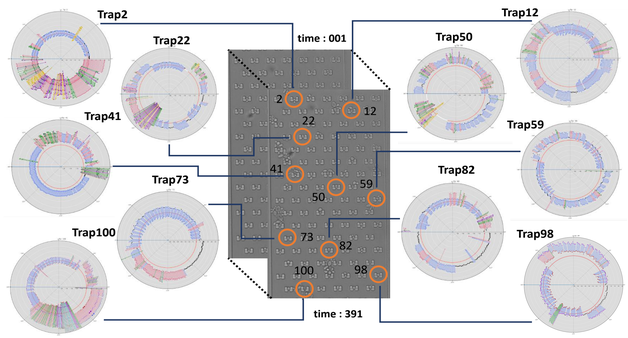
(f) Representation of steady cells in the up region (dark red color) and developing cells in the up region (purple color)

(g) Two cells are in a steady situation for some time, an indication of senescent cells

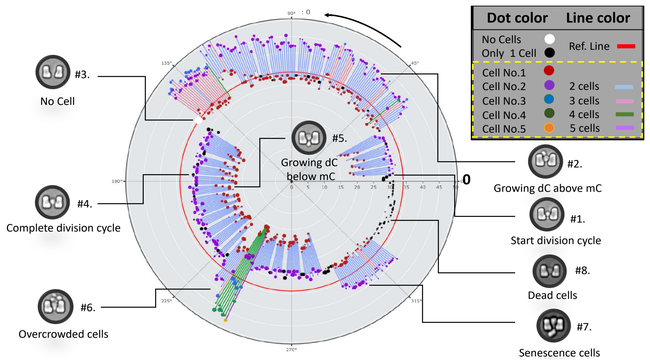
(h) The representation of a cell that has been dead for some time

Results

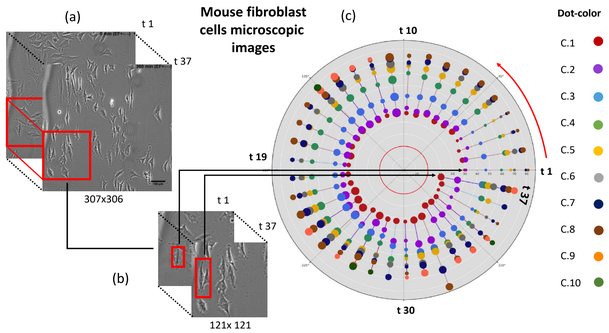
Time-lapse full microfluidic images with µPolar plots for Trap No. 2, 12, 22, 41, 50, 59, 73, 82, 98 and 100.



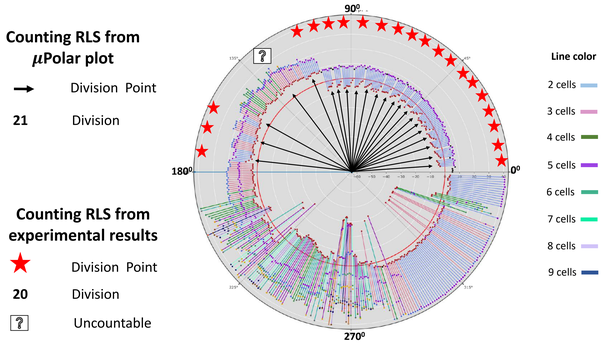
A µPolar plot with cell area representation and cell tracking for Trap No. 98.The line color represents the number of cells at each time-point, and dot color represents a color tag for individual cell at each time-point, respectively.



µPolar plot for migrating mouse fibroblast cells in time-lapse microscopic images. Cell areas are represented by dots with variable diameters, and the same cell objects in each timepoint are marked by a unique color.



Identifying RLS measurements from µPolar plot for Trap No. 1. The black arrows indicate the potential number of yeast cell division time-points from µPolar plot estimating 22 divisions. The red stars indicate the 21 cell division events identified from the experimental results.



Discussion

Overall, µPolar is a useful tool for visualizing cell migration, cell monitoring, and estimating cell division events from time-lapse microscopic images. The µPolar interpretation of cell division at each time-point can facilitate lifespan estimation in aging studies. The comparison between yeast cell time-lapse microfluidics images and mouse fibroblast cell time-lapse microscopic images demonstrates that µPolar can be a general tool for visualizing time-lapse images. Visualizing cell division events can offer biological insights. For instance, in the microfluidic images, when yeast cells become older, interestingly, the cell division cycle becomes longer. It can also be seen that there is a relationship between cell size and cell division time-length. Similarly, visualizing and tracking mouse fibroblast cells show how cell sizes change during their migration.

Conclusion

- We developed an R package based on a circular plotting method, μPolar, to visualize cell movements and cellular division events at hundreds of time points.
- Our method is 2D, interactive and easy to use.
- We demonstrated the utility of our method to describe the events of dividing yeast cells and migrating mouse fibroblast cells.

Overall, μPolar visualization tool could be applied to other types of microfluidic devices and time-lapse microscopic imaging experiments.

Acknowledgements

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